Lecture 3: Arithmetic & Logic Operations (MCQs)

1. Mathematically, each Logical Shift to the Left (LSL) _________ the number.
a) Divides by 2 b) Multiplies by 2 c) Increments by 1 d) Inverts the sign

• **Answer:** b) Multiplies by 2

2. Mathematically, each Logical Shift to the Right (LSR) _________ the number.
a) Divides by 2 b) Multiplies by 2 c) Decrements by 1 d) Inverts the sign

• **Answer:** a) Divides by 2

3. In 2's Complement Signed Multiplication, you must _________ before multiplying.
a) Invert all bits b) Add 1 to the LSB c) Sign extend operands to the product bit width d) Convert to Unsigned

• **Answer:** c) Sign extend operands to the product bit width

4. If you multiply two 4-bit numbers (e.g., -7 * 6), the result should typically be stored in _________ bits to avoid overflow.
a) 4 b) 6 c) 8 d) 16

• **Answer:** c) 8

5. What is the result of shifting 0100 one bit to the right?
a) 1000 b) 0010 c) 0001 d) 0110

• **Answer:** b) 0010

6. When performing signed multiplication of 101 (-3) and 011 (3) without sign extension, the result is incorrect because _________.
a) The carry is lost
b) The sign bit is not preserved correctly in the product
c) It causes a divide by zero error
d) The operands are too large

• **Answer:** b) The sign bit is not preserved correctly in the product

7. In the division example 1001 / 11, the quotient (result) is _________.
a) 0010 b) 0011 c) 0100 d) 1100

• **Answer:** b) 0011 (Decimal: 3)

8. The operation that shifts bits where the bit exiting one end enters the other end is called _________.
a) Logical Shift b) Arithmetic Shift c) Cyclic Shift (Rotate) d) Truncation

• **Answer:** c) Cyclic Shift (Rotate)

9. In a Cyclic Shift Left through Carry, the MSB moves to ________.
a) The LSB position b) The Carry Flag (C) c) The Overflow Flag (V) d) It is discarded

- **Answer:** b) The Carry Flag (C)

10. If the register contains 0111 and a Logical Shift Left (LSL) is executed, the new value is ________.
a) 0011 b) 1110 c) 1111 d) 1000

- **Answer:** b) 1110

11. Referencing the previous question (0111 -> LSL -> 1110), what is the value of the Carry (C) flag after the shift?
a) 0 b) 1 c) Undefined d) High Impedance

- **Answer:** a) 0 (Because the bit shifted out from MSB was 0)

12. Referencing the shift 0111 -> 1110, what is the value of the Overflow (V) flag if this represents a signed number?
a) 0 b) 1 c) No change d) It depends on the previous instruction

- **Answer:** b) 1 (Because the sign bit changed from 0 to 1, indicating overflow)

13. To calculate -L * 2 for a binary number, the correct sequence of operations is:
a) Shift Left -> 2's Complement
b) 2's Complement -> Shift Left
c) Shift Right -> 2's Complement
d) 1's Complement -> Shift Right

- **Answer:** b) 2's Complement -> Shift Left

14. If L = 010011 (19), what is the binary result of -L (in 6 bits)?
a) 101100 b) 101101 c) 110011 d) 001101

- **Answer:** b) 101101 (Invert bits 101100 + 1)

15. In binary division, if the divisor cannot be subtracted from the current portion of the dividend, you must ________.
a) Put a 1 in the quotient
b) Put a 0 in the quotient and include the next bit
c) Stop the operation
d) Set the Overflow flag

- **Answer:** b) Put a 0 in the quotient and include the next bit

16. Which operation effectively performs Multiplication by 4?
a) Shift Right by 2 b) Shift Left by 1 c) Shift Left by 2 d) Rotate Right by 2

- **Answer:** c) Shift Left by 2 (x2 then x2)

17. When multiplying by hand (Paper & Pencil method) for signed numbers, it is often easier to ________.

a) Use 1's complement

b) Multiply positive values then adjust the sign

c) Ignore the sign bit completely

d) Always add 1 to the result

• **Answer:** b) Multiply positive values then adjust the sign

18. In a 4-bit Signed Multiplication of 1111 (-1) and 0001 (1), the 8-bit result should be: a) 00000001 b) 11111111 c) 10000001 d) 00001111

• **Answer:** b) 11111111 (Result is -1)

19. Cyclic Shift Right on 0011 (without carry) results in:

a) 0001 b) 1001 c) 1000 d) 0110

• **Answer:** b) 1001 (The LSB 1 rotates around to the MSB)

20. What flag usually holds the bit that was last shifted out?

a) Z (Zero) b) N (Negative) c) C (Carry) d) P (Parity)

• **Answer:** c) C (Carry)

21. For the hexadecimal operation 4A shifted left by 1 bit, the result is:

a) 25 b) 94 c) 84 d) 4B

• **Answer:** b) 94 (4A is 0100 1010 -> Shift Left -> 1001 0100 -> 94)

22. If 1001 is divided by 11 using binary long division, the remainder is ________.

a) 0 b) 1 c) 10 d) 11

• **Answer:** a) 0 (Since 3 divides 9 perfectly)

23. Which instruction is likely used to inspect specific bits of a register without changing them? a) Shift b) Rotate c) Add d) Load

• **Answer:** b) Rotate (Bits are preserved, just moved)

24. Sign Extension means filling the upper bits of the wider format with ________.

a) Zeros b) Ones c) The value of the MSB (Sign Bit) d) The value of the LSB

• **Answer:** c) The value of the MSB (Sign Bit)

25. If C=1 and V=1 before a shift, and after the shift C=0 and V=0, this implies:

a) No overflow occurred in the last operation

b) The result is negative

c) The carry is set d) An error occurred

• **Answer:** a) No overflow occurred in the last operation

26. If a 4-bit register 1000 (-8) is shifted left (LSL), what happens?
a) Result is 0000, Overflow occurs
b) Result is 1000, No change
c) Result is 0001, Sign changes
d) Result is 1001

• **Answer:** a) Result is 0000, Overflow occurs (Because -16 cannot fit in 4 bits)

27. To divide a number by 4 using shifts, you would apply:
a) 1 Right Shift
b) 2 Right Shifts
c) 2 Left Shifts
d) 4 Right Shifts

• **Answer:** b) 2 Right Shifts

28. In the multiplication example 1010 (-6) x 0111 (7), the intermediate partial products are shifted to the _________.
a) Right b) Left c) Are not shifted d) Rotated

• **Answer:** b) Left (Standard multiplication shift direction)

29. The binary representation of 15 is 1111. If shifted left by 1 bit, it becomes 11110, which equals _________.
a) 16 b) 30 c) 32 d) 14

• **Answer:** b) 30

30. Which of the following is TRUE about signed multiplication in 2's complement?
a) You can simply multiply the bits like unsigned numbers.
b) You must stop after the first 4 bits.
c) Overflow is impossible.
d) You must sign-extend operands to avoid corrupting the sign.

• **Answer:** d) You must sign-extend operands to avoid corrupting the sign